



UG Program in Electronics and Communication (Advanced Communication Technology)

Experiment No. 6

Generation (and detection) of Binary FSK

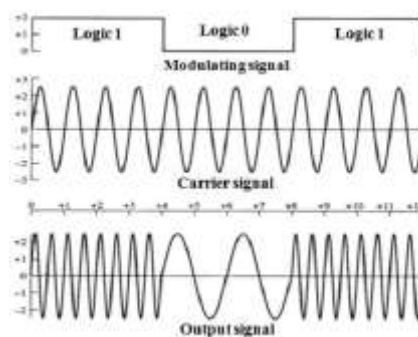
6.1 Aim: To study the Binary Frequency Shift Keying Generation (and detection) technique and plot the modulated and demodulated waveforms.

6.2 Course Outcome: Compare the performances of different digital modulation techniques

6.3 Learning Objectives: Students will be able

- To analyze and investigate data sets, summarize their main characteristics with data visualization.
- Apply Machine Learning techniques to predict disease risk from Patient data for Parkinson's Disease.

6.4 Related Theory:



BFSK Modulation Waveforms

In Frequency shift keying, the carrier frequency is shifted (i.e., from one frequency to another) corresponding to the digital modulating signal. If the higher frequency is used to represent a data '1' & lower frequency a data '0', the resulting FSK waveform appears. Thus



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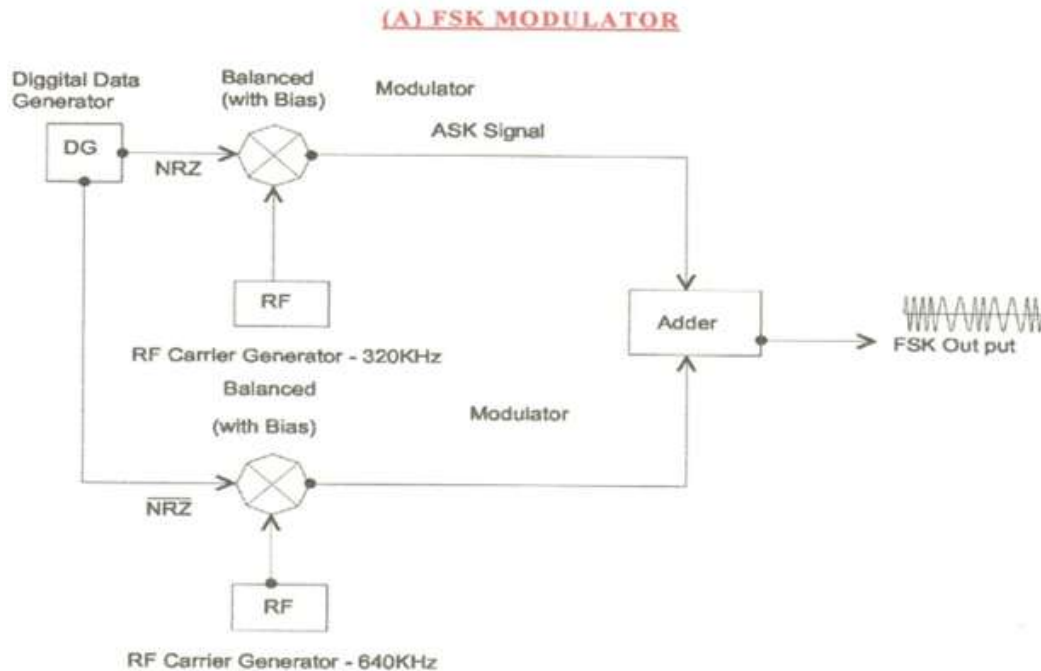
Data =1 High Frequency

Data =0 Low Frequency

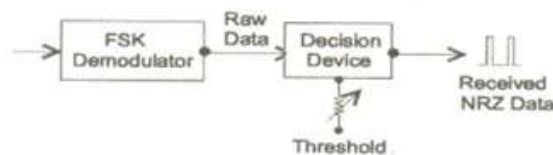
It is also represented as a sum of two ASK signals. The two carriers have different frequencies & the digital data is inverted. The demodulation of FSK can be carried out by a PLL. As known, the PLL tries to 'lock' the input frequency. It achieves this by generating corresponding O/P voltage to be fed to the VCO, if any frequency deviation at its I/P is encountered. Thus, the PLL detector follows the frequency changes and generates proportional O/P voltage. The O/P voltage from PLL contains the carrier components. Therefore, to remove this, the signal is passed through Low Pass Filter. The resulting wave is too rounded to be used for digital data processing. Also, the amplitude level may be very low due to channel attenuation.

A block diagram of BFSK modulation demodulation system consist of following sections

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(B) FSK DEMODULATOR (PLL DETECTOR)



The Generation & Reception of FSK consists of the following sections.

1. Digital Data Generator section
2. ASK modulator using Balanced Modulator.
3. Carrier Generator
4. Adder section.
5. FSK demodulator using PLL detector



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6. Data Squarer (Comparator) Section
7. Power supply section

Digital Data Generator section:

To generate FSK signal input digital modulating signal is required. This section generates digital data signal. It consists of many digital ICs. The carrier clock of 128 MHz is generated by IC 74HC04, this carrier clock frequency is divided by 16 to produce bit clock (Bk) of 80 KHz. This Bk frequency is divided by 8 to generate Word clock (Wk) of 10 KHz. Then a word of 8-bit data is generated using IC 74166 and eight data toggle switches by pressing different toggle switches different data word can be generated. This data word is called as NRZ data. The NRZ data is then given to the ASK modulator.

ASK modulator using Balanced Modulator: -

IC 1496 is used as a balanced modulator. This IC has two inputs as it works as a balanced modulator. The modulating signal is connected at pin 1 through buffer transistor BC5488. The second input can be connected at pin 4 through buffer transistor BC548B. The RF carrier signal is connected at pin 8 through a coupling capacitor from the RF carrier oscillator section. The modulated outputs are available at pin 12 and 6 of this IC, which are then balanced amplified by transistor pairs - BC548B and BC558B. The final balanced modulated output is available at the O/P terminal. 100K preset is used to balance the carrier signal while 1K preset is used to balance input audio signal. Output 1K preset is used to adjust output zero DC level. Output 2K2 preset is used to vary output level. The outputs are two ASK Modulated signals. There are two ASK modulators: one is used with NRZ data and other is used with NRZ data.

RF Carrier Generator section: -

The function of this section is to generate phase synchronized carrier signal with respect to modulating digital NRZ input. It converts 640 KHz and 320 KHz TTL clock signal into 640 KHz and 320 KHz sine wave carrier signal using piezoelectric of 100 μ H coil and two 100 pF capacitors. These signals are given as carrier signal; to balanced modulators as one input.

Adder section: -

This section adds two input signals with unity gain. This is used to add to ASK signals to convert into FSK signal.

FSK demodulator: -



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This circuit is based on PLL made of IC 4046. The free running frequency of PLL is set at 320KHz value. Hence when FSK signal having two frequency 320KHz and 640KHz is given to its input, it produced demodulated signal locked to 320KHz. Hence output is raw data signal. This signal is then given to squarer circuit.

Data Squarer (Comparator) section: -

This section converts raw data output of Frequency shift keying demodulator into pure recovered data using comparator IC 311.

Power supply section: -

The regulated power supply is used for different supply voltages. Following output D.C. Voltages are required to operate FSK Modulation demodulation system:

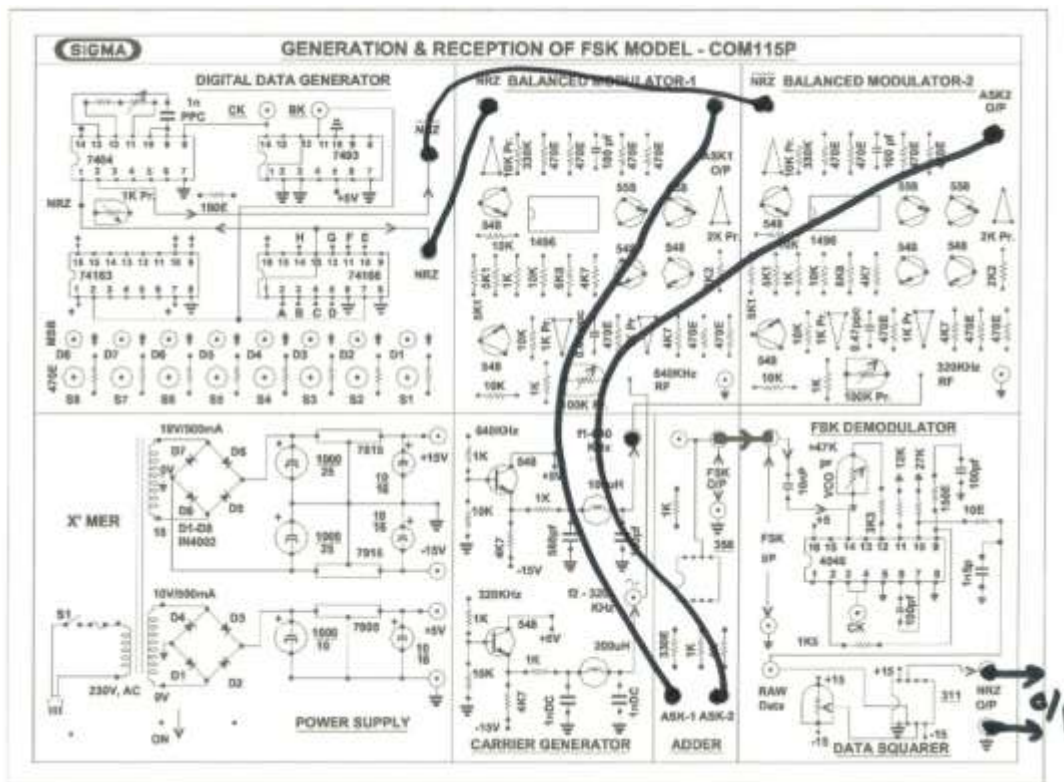
+15V 250 mA, -15V 250 mA, + 5V 250 mA.

Three terminal regulators are used for different output voltages i.e., IC 7805 for +5V, IC 7815 for

+15V, and IC 7415 for -15V. These ICs are supplied different dc input voltages by two Bridge rectifiers consisting of DI-D4 and D5-D8 & two 1000125 EC and 1000/10 EC. The capacitors at each input & each output are for filtering purpose

Circuit Details:

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Procedure:

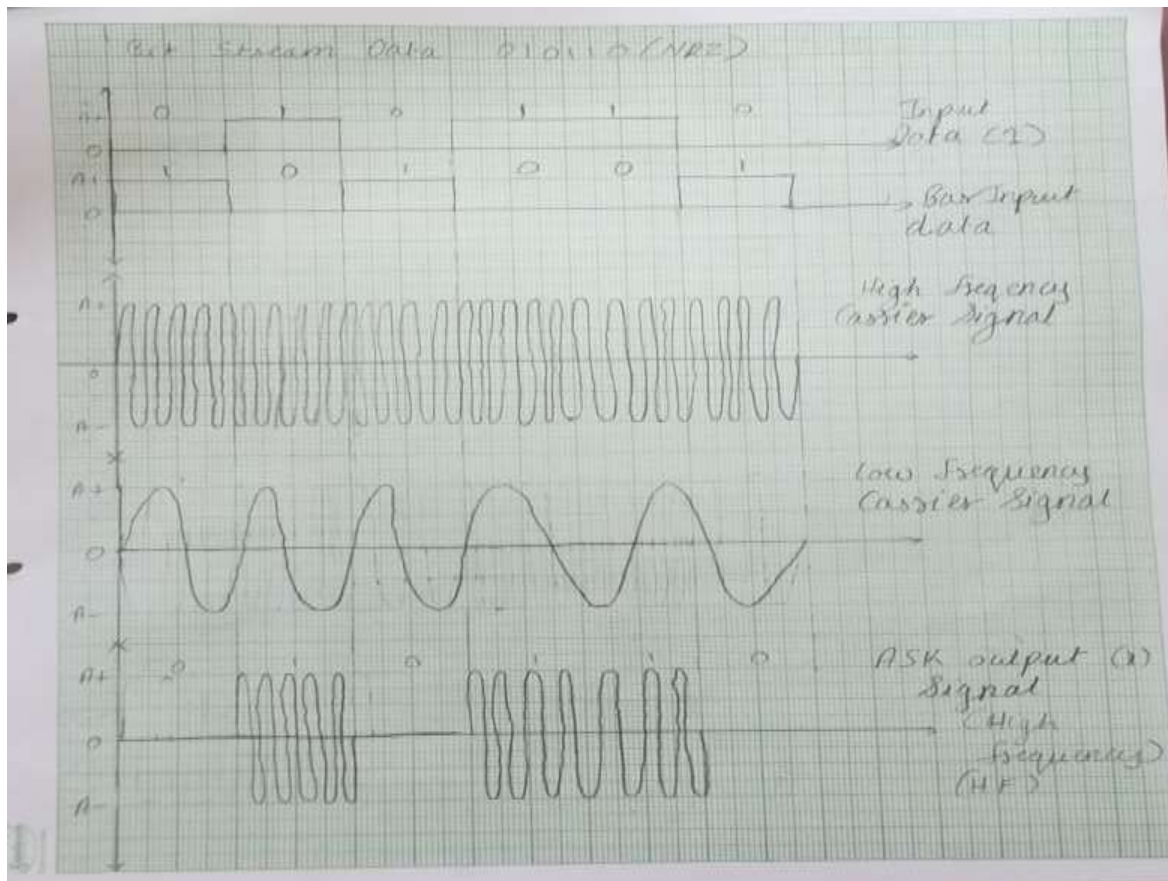
1. Connect CRO Channel-I at Carrier Clock (CI) socket and observe it
2. Connect CRO Channel-I at Bit Clock (Bk) socket and observe it
3. Toggle the switches to give data input and connect CRO Channel-I at NRZ DATA socket and CRO Channel-I at NRZ DATA socket and observe it.
4. Connect CRO channel at RF Carrier socket (in carrier generator section) for frequency 320KHz and 640KHz, observe the waveforms on CRO for these two frequencies.



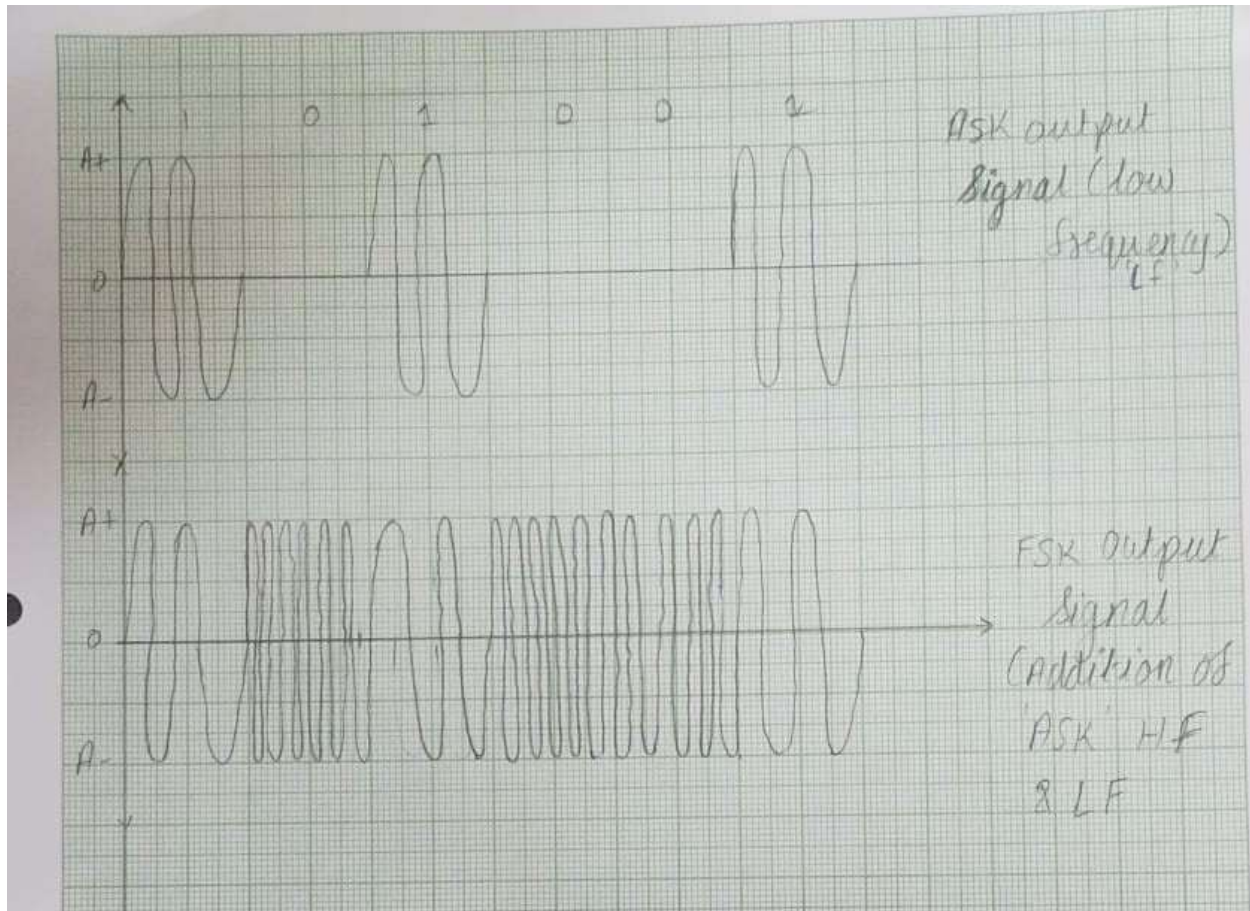
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5. Connect CRO channel-1 at NRZ DATA (NRZ) socket and channel 2 at the output of balance modulator-I Observe ASK output measuring its amplitude and frequency.
6. Similarly connect CRO channel 2 at the output of balance modulator 2. Observe ASK output.
7. Connect CRO channel 2 at output of adder and observe FSK output measuring its amplitude and frequency.
8. Observe recovered raw data signal at output of FSK demodulator and observe received pure NRZ data at the output data square.

6.5 Output WAVEFORMS:



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6.6 Conclusion:

In this experiment Binary Frequency Shift Keying (BFSK) modulation and demodulation were successfully studied. The FSK signal was generated by combining two ASK signals of different frequencies. Demodulation using a PLL accurately recovered the original NRZ data.

Thus, the experiment verifies that BFSK provides reliable digital communication with better noise immunity than ASK.



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6.7 Post lab questions:

1. What is Frequency Shift Keying (FSK)?

Ans - FSK is a digital modulation technique in which the carrier frequency is changed according to the binary input data.

2. What is the bandwidth requirement for FSK signals?

Ans - FSK requires more bandwidth than ASK but less than PSK, depending on the frequency separation.

3. What is the main advantage of FSK over ASK?

Ans - FSK is less affected by noise because information is carried by frequency rather than amplitude.

4. Name one application of FSK in communication systems.

Ans - FSK is used in modems, Bluetooth, and wireless data transmission systems.